Executable Evidence Management for CAD-Guided Aircraft Inspection

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Abstract

Engineering changes alter the evidence supporting inspection outputs. We develop executable operating contracts that connect CAD-guided aircraft inspection requirements to numerical services and reusable statistical statements. A typed representation and unit-aware compiler link task events, processing policies, producer dependencies and numerical versions. Requirement containment supports evidence reuse, registered dependency signatures support selective renewal, and persistent execution history connects evidence to the output produced. A reuse-first route with exact-event fallback retains all 59 directly supported request contexts under fixed numerical policies. Selected evidence preparation requires 63,488 event–record evaluations rather than 73,728, a 13.89% reduction. A controlled camera campaign recomputes 2,560 paired updates across 320 configurations, covering pose, focal length, exposure and brightness variation. Four conditional coupling classes activate in 747 cases; output-difference auditing identifies each activated update while accepting all 1,280 declared updates. The underlying calibration and test studies include 13,824 localization episodes, 36 continuous-route inputs and 12,800 image-task inputs. The framework coordinates evidence retention, withdrawal and restoration across localization, image acquisition, CAD association and reporting.

Keywords: Executable operating contracts, Evidence reuse, Engineering change management, Risk calibration, Aircraft inspection

1. Introduction

An aircraft inspection image becomes useful engineering information when its location, acquisition quality and reporting status are known. A robot must place the image on the aircraft model, acquire the required surface region and deliver the result within the operating deadline. These requirements connect geometric estimation, camera settings and information processing. They also change during operation: an inspector can request a larger region, an updated decoder can alter image association, and a later pose estimate can replace the version available at capture.

The resulting challenge is to maintain the evidence supporting an inspection output through those changes. Repeating every calibration is wasteful when a change leaves the requested quantities intact. Retaining every previous statement is unsafe when the producer or its applicable population has changed. The engineering decision therefore concerns a specific request: which evidence remains valid, which statement must be withdrawn, and which computation is needed to restore service?

Existing methods address important parts of this problem. Engineering knowledge models relate decisions to requirements and dependencies [1, 2, 3]; anytime computation connects estimation quality to processing time [4, 5]; risk-controlled selection supplies statistical bounds for specified programs [6]. Connecting these operations requires an explicit relationship between a consumer requirement, the producer that computes its quantities and the numerical version used by the output. A component dependency alone does not identify the statistical event certified by a calibration result.

We develop executable operating contracts for CAD-guided aircraft inspection. The contract represents a request in engineering units, links its failure event to applicable evidence, and binds the resulting plan to a committed computation policy. Requirement containment allows an existing bound to support a changed request. Requested dependency signatures identify evidence unaffected by a registered producer change.

Ordered numerical versions and persistent execution history determine whether that evidence can accompany the output. Figure 1 locates these decisions in the acquisition and CAD-association workflow; Fig. 2 gives the executable architecture.

The contribution is an evidence-management capability across the inspection pipeline. We provide a typed representation and request compiler, admission and transfer rules, and a reuse-first route with exact-event renewal. The evaluation connects request support, proof-preparation work and actual producer changes. On fixed numerical policies, the route preserves all 59 directly supported contexts and reduces selected evidence evaluations from 73,728 to 63,488. A controlled camera campaign evaluates 2,560 paired updates across 320 configurations, while population calibration and numerical studies characterize the underlying services.

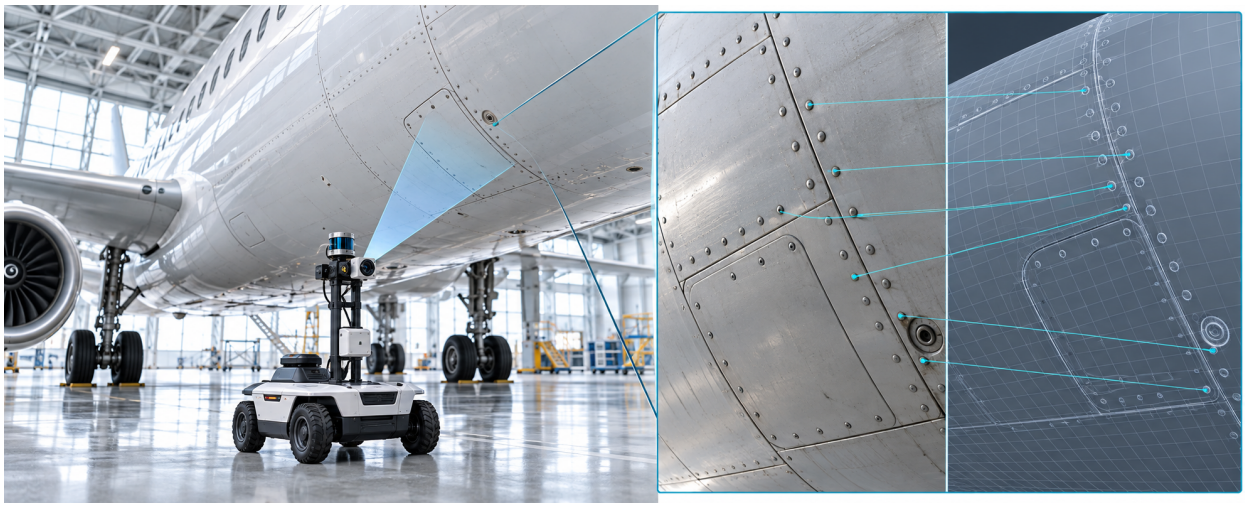


Figure 1: Conceptual view of CAD-guided inspection. Robot localization places acquired surface information in the aircraft model; image features connect the camera view to model addresses. Task requirements govern the quality and evidence needed for this association.

2. Related work

2.1. Engineering knowledge and change operations

Process knowledge and decision knowledge graphs organize engineering choices and their supporting context [1, 2]. Semantic change propagation makes component relations available for impact analysis [3, 7], while inspection ontologies connect operating conditions to robot requirements [8]. Their contribution is assessed through engineering operations: representational coverage and queries [9], or executable conversion and downstream model fidelity [10]. Our corresponding operation is the continued applicability of a statistical statement after a task or provider changes. It requires both the quantity's implemented dependency projection and the failure event certified by the original card. A component-impact relation becomes actionable when it identifies the certified event, applicable population and permissible evidence transfer. These links define the engineering operation studied here.

2.2. Computation contracts and source revisions

Metareasoning and anytime algorithms allocate computation according to its expected value [11, 12, 13]. Pant et al. connect computational contracts to estimation and control [4]; Aldana-López et al. study stability-preserving perception scheduling [5]. We retain ICP and joint-state estimation [14, 15, 16], using complete-event calibration alongside approximate curvature covariance [17, 18, 19]. The interface tracks correlated checkpoints as revisions of one source, then links the version used at capture to the requested evidence. This source identity is carried through replacement and causal motion replay.

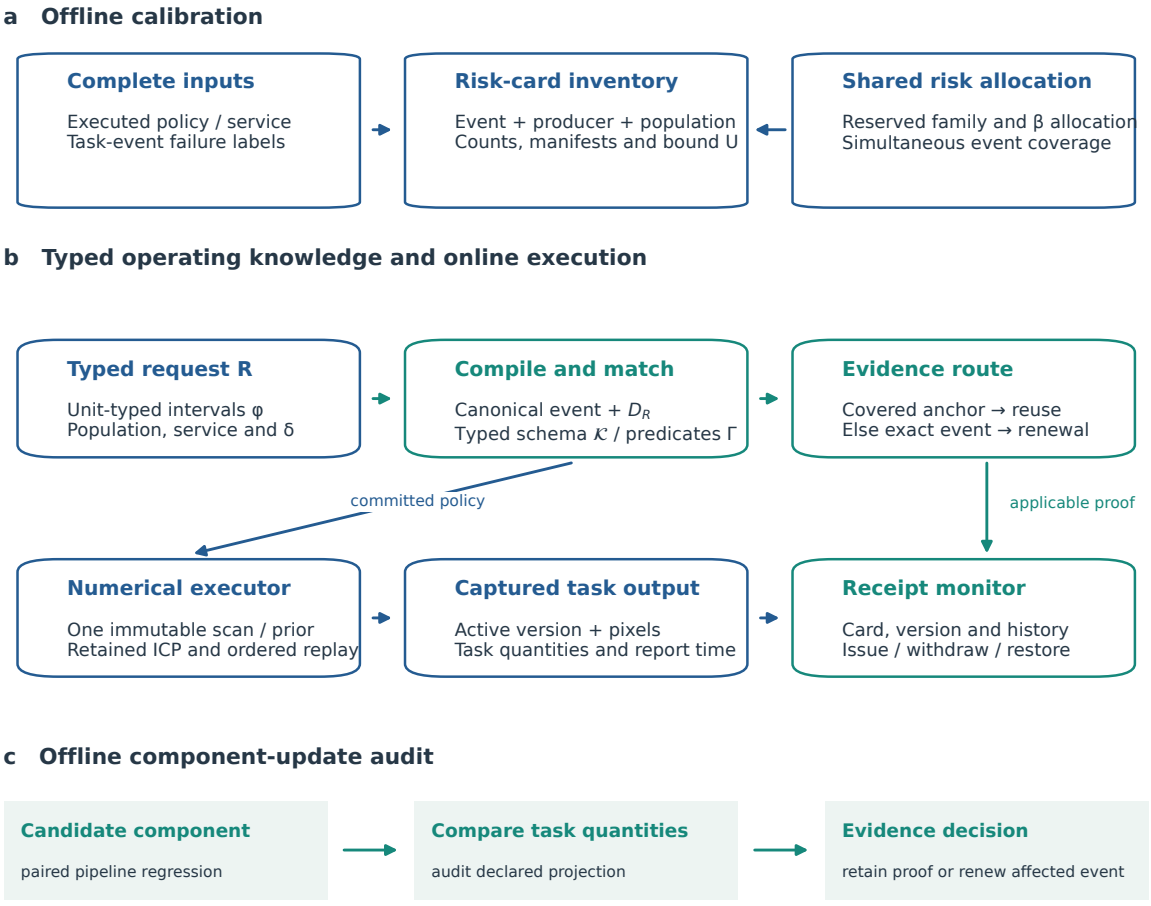


Figure 2: Architecture of the executable evidence interface. (a) Calibration records supply event-specific cards; typed requirements compile into a committed policy and evidence plan. (b) Source-consistent numerical execution supplies capture-time quantities and ordered versions. (c) Producer auditing, dependency projections and source-history checks govern reuse, direct fallback and issuance. The metamodel links requirements to evidence and execution; component auditing occurs during update registration.

2.3. Risk selection and evidence lifetime

Assumption–guarantee contracts formalize component obligations [20]; binomial inversion and risk-controlled selection calibrate specified configurations [21, 6, 22]. We use simultaneous finite-family coverage to support before-acquisition policy and anchor selection, then extend each anchor through explicit containment and requested-function invariance. Runtime assurance supplies assumption monitoring and controller switching [23]. Here the monitored action is evidence issuance, with a source-scoped history latch that persists through plan reconstruction while numerical installation continues through its original interface.

Table 1 identifies the links introduced by the operating contract relative to these established operations. The central contribution is an executable evidence lifecycle: task/event identity, producer applicability and numerical source history jointly govern admission, retention, withdrawal and restoration. The evaluation measures those operations directly through matched preparation counts and change decisions.

Table 1: Positioning by engineering operation. Each row maps an established research operation to the executable relation evaluated here; it is a conceptual comparison of scope.

Established operation	Executable relation introduced here	Evaluation endpoint
Knowledge and change reasoning [1, 2, 3]	Requested quantities link provider changes to specific certified events	Affected statements withdrawn; applicable statements retained and restored
Anytime estimation and scheduling [4, 5]	Immutable-source revisions link a committed policy to the version consumed at capture	Source trace, localization trade-offs and complete-route availability
Risk-controlled configuration selection [6, 22]	Containment and invariant requested projections link new requests to existing risk cards	Unique request support and preparation work on matched contexts
Runtime assumption monitoring [23]	Persistent source history links attachment attempts to evidence status	Invalid-packet refusal, gate reconstruction and trusted-scope recovery

3. Executable task and evidence model

3.1. Typed task–source–evidence representation

The operating knowledge structure is

$$\mathcal{K} = (\mathcal{V}, \tau, \mathcal{R}, \sigma, \Gamma), \tag{1}$$

where entities $\mathcal{V}$ have types τ, relations $\mathcal{R}$, immutable specifications σ and executable predicates Γ. The core types are request, event, policy, source version, producer, quantity, provider, card, plan and receipt. Relations have explicit domains and ranges: a request *requires* an event, a card *bounds* that event and *supports* a plan, a plan *commits* a request and *selects* a policy, and a receipt *uses* a source version. Quantities *depend on* providers. A *revises* edge requires the same immutable source key and a strictly increasing sequence. The supplement gives the complete relation schema and executable instance checks.

A request $R = (g, a, \varphi, \delta)$ fixes population, service, a predicate conjunction and risk allowance before sensing. Its grammar is

$$\varphi ::= \mathrm{Interval}(m, u, \ell, h) \mid \varphi \wedge \varphi, \tag{2}$$

where m identifies a registered quantity, u its engineering unit and omitted bounds represent unbounded sides. The compiler checks dimensions, converts bounds to base units, intersects repeated intervals, rejects contradictory specifications and sorts quantities into a canonical event identity. It returns the required event and the union of its provider dependencies. Nonfinite required quantities fail the compiled event. The executable compiler reproduces all 24 declared task/variant events, including their dependency projections, and checks invalid units, bounds and relation types.

The implementation records quantity dependencies in a versioned manifest derived from data flow. Component revisions pass through an offline regression gate, where common inputs are rerun and observed output changes are compared with the proposed dependency declarations. An undeclared change blocks registration. During request processing, the interface compares registered dependency signatures, numerical versions and execution history; it does not rerun the renderer or solver for every request. The representation supplies a typed execution and evidence interface, with predicates for unit validity, policy/population matching, containment, dependency projection, active source version and persistent history. The admission rule is

$$\begin{aligned} &\mathrm{ValidCard}(A) \wedge \mathrm{Match}(A, R, \pi) \wedge \mathrm{Cover}(A, R) \\ &\quad \wedge \mathrm{SameProjection}_{D_R}(p, p_A) \wedge U_A \le \delta \implies \mathrm{Admissible}(\pi, A, R). \end{aligned} \tag{3}$$

The following sections define containment, transfer and source-history semantics. Runtime issuance additionally requires a valid source identity and integrity check, the active ordered version and a valid persistent history. These predicates determine inspectable admission, withdrawal and restoration actions.

Table 2: Typed operating records and their executable roles. Events, quantities and provider identifiers supply the nested metamodel.

Record	Required specification	Executable decision
Request/event	Population, service, unit-typed intervals, allowance	Compile and check the required event
Policy/service	Effective program, action rule, budget, load, deadline	Commit a policy before sensing
Source/version	Immutable scan/prior key, sequence, payload, integrity check	Replace the prior contribution and replay motion
Producer/quantity	Quantity functions and declared provider snapshot	Audit changes and compare requested projections
Evidence card	Event, program, population, source manifests, n, k, U	Validate the bound and containment witness
Plan/receipt	Request, policy, card, source version, persistent history	Issue, withdraw or restore applicable evidence

3.2. Source-consistent numerical execution

Localization provides the coordinate transformation that places a camera observation on the aircraft model. The estimator jointly tracks robot pose, motion bias and map pose. For state $x = (t, R, b, m_t, m_R)$, the airframe-relative position error is

$$e_p = \widehat{m}_R^\top(\widehat{t} - \widehat{m}_t) - (m_R^\star)^\top(t^\star - m_t^\star). \tag{4}$$

Measured motion propagates the joint state and its 18-dimensional covariance using an $SE(3)$ linearization [16].

A rolling scan supplies several progressively refined registration estimates. They are revisions of one observation, so using them as independent measurements would count the same scan repeatedly. We retain the scan and its original prior, continue the solve, and replace the active contribution when a newer eligible estimate arrives. Point-to-plane registration minimizes

$$\Phi_s(x) = d^\top (P_s^-)^{-1} d + \sigma^{-2}\left[\sum_{i\in\mathcal{I}(x)} \rho(r_i(x)) + (n - |\mathcal{I}(x)|)\rho(r_{\text{out}})\right], \quad d = x \boxminus x_s^-. \tag{5}$$

The unmatched-point term retains difficult correspondences. Motion compensation and voxel filtering produce up to 1,500 frontend points; the primary solver uses at most 96 deterministically thinned points from the local visible patch. Checkpoints at iterations 1, 2, 4, 8, 16 and 25, or termination, permit useful intermediate delivery. Supplementary methods give the robust loss, covariance and acceptance criteria.

When version j arrives at time t_d, its source contribution is replaced and already acquired motion is replayed:

$$(x_{t_d}^{(j)}, P_{t_d}^{(j)}) = \mathcal{R}_{s\to t_d}(x_s^{(j)}, P_s^{(j)}; u_{s+1:t_d}). \tag{6}$$

Capture consumes the version available at its timestamp. This ordering matters to evidence management: a later improvement cannot change which estimate produced an earlier image association. Under a retained solve with no interleaved independent update, final-version replacement gives the same state as delivering that version alone at the same time. Figure 3 shows how an executed source, a captured observation and a changed request remain connected.

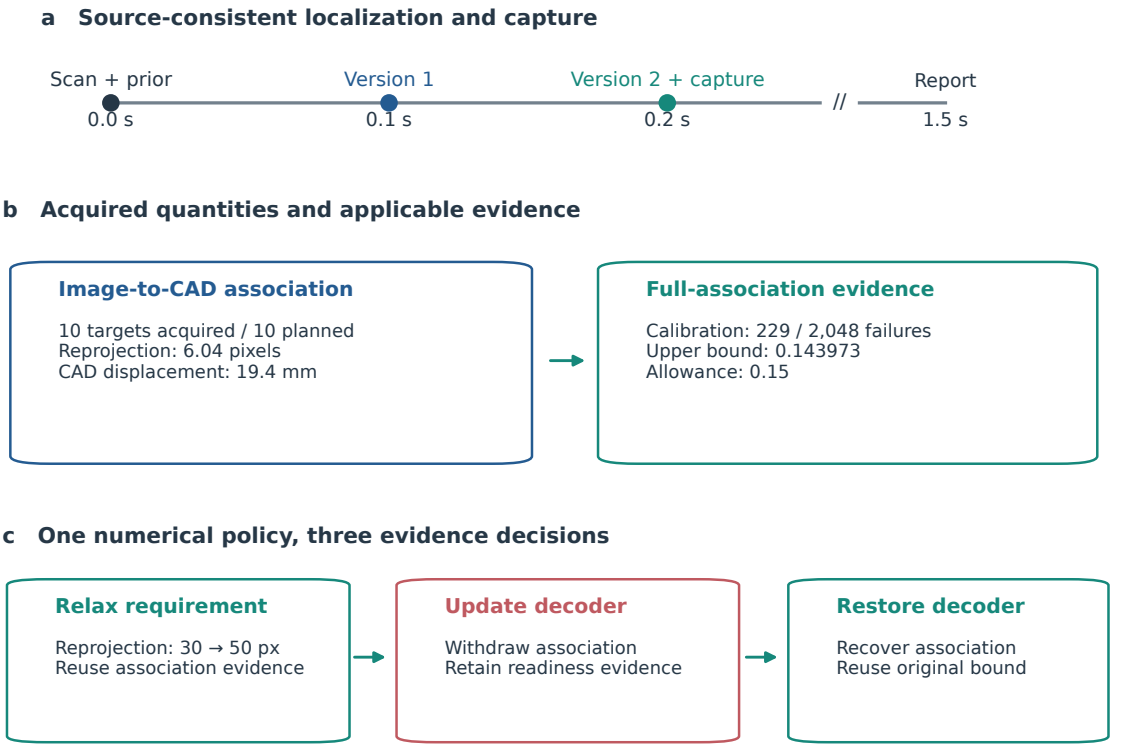


Figure 3: One executed localization source and its evidence lifecycle. Ordered installations determine the version used at capture. The measured association quantities support a calibrated statement; requirement relaxation, decoder change and restoration lead to reuse, withdrawal and recovery under the same numerical policy.

3.3. Events, policies and computation budgets

For a four-second episode, integrated position loss is $L_\pi = \int_0^4 \|e_p^\pi(t)\|^2 dt$, evaluated by a 0.1-s right-endpoint sum. At study threshold $\epsilon = 0.10$ m, define

$$E_{\mathrm{tr}} = \{\max_{t\in\{0.5,0.6,\ldots,4.0\}} \|e_p(t)\| > \epsilon\}, \tag{7}$$

$$E_{\mathrm{obs}} = \{\text{no positive source version at } D\} \cup \{\|e_p(D)\| > \epsilon\}, \quad E_{\mathrm{un}} = E_{\mathrm{tr}} \cup E_{\mathrm{obs}}. \tag{8}$$

Same-tick ordered installation and rollback precede assessment. Trajectory, observation, both-marginal and union requests use these distinct events. A union request uses its directly evaluated OR event; two marginal bounds yield only their summed union bound.

The retained CART and random-forest predictors share a 1,120-source development set, split into 926 training and 194 validation sources. CART has depth seven and minimum leaf size 20; the forest averages 64 depth-eight trees with minimum leaf size 12. Causal features describe uncertainty, information, residuals, steps, iteration, pair count, displacement and declared noise. Predictors estimate terminal MSE $\widehat{v}_{ja}$, remaining work $\widehat{c}_{ja}$ and a development-event frequency $\widehat{r}_{ja}$ for reachable cap a. The endpoint heuristic selects

$$\widehat{a}_j = \arg\min_{a\in\mathcal{A}_j} \{T\max(\widehat{v}_{ja}, 0) + \lambda\max(\widehat{c}_{ja}, 0) + \omega\,\mathrm{clip}(\widehat{r}_{ja}, 0, 1)\}, \tag{9}$$

with $T = 4$ s denoting the episode horizon, $\lambda \in \{0.3, 1, 3\}$ m^2 and $\omega = 0.002$ m^{2}s. MSE comparisons also guide publication of eligible candidates. These forecasts are scheduling heuristics; event calibration concerns the complete executed wrapper.

The localization catalogue contains 22 policies: matched CART/RF variants at three prices, fixed caps, final convergence, residual and randomized controls, the earlier CART scheduler and time-only interventions. We call the price-one time/profile-screened policy Contract-CART; its archived identifier is KG1. Earlier CART denotes the preceding scheduler, stored as OldRAVC in the reproducibility tables. FlatCART uses the same predictor; time-only and contract variants add deadline and profile predicates in sequence. Every policy receives the common next-stage budget reservation, acquisition and source-replay service. Hard-budget interruption is therefore part of each effective policy, including Fixed2.

The stage budget B covers source identity and integrity checking, registration, features, forecasting, rules and publication. Total modeled work W additionally includes the common front end, receiver and stream. The fixed operation clock assigns CPU-equivalent costs to actual counted operations, then applies the declared load factor to delivery scheduling. Native CPU is a separate executed-work measurement. Selection uses the dimensionally consistent calibration objective $\overline{L} + \lambda_{\mathrm{sel}}\overline{W}$, with $\lambda_{\mathrm{sel}} = 1\ \mathrm{m}^2$. Supplementary methods give the installation-clock equation and the boundaries of component and whole-route CPU.

3.4. Calibration, admission and evidence lifetime

A typed contract records the source, numerical version, policy, population, service, requested event and their dependencies. Before opening the scan, a request fixes its event set and risk allowance δ. For a fixed event with k failures among n iid complete calibration inputs, the stored one-sided bound is

$$U(k,n) = \begin{cases} 1, & k = n, \\ \mathrm{Beta}^{-1}(1-\beta/K; k+1, n-k), & k < n. \end{cases} \tag{10}$$

With $\beta = 0.05$ and localization family $K = 22\times2\times4\times3 = 528$, a union bound gives simultaneous risk coverage of at least $1-\beta$. Dependence between policies or events on the same source is allowed. Calibration-dependent selection can use the resulting bounds because they hold for the entire fixed family on one confidence event.

An admitted policy matches the request's population, service and dependencies and satisfies $U \le \delta$ for every required event. The selector minimizes the same calibration loss–work objective and commits before sensing. An empty admissible catalogue refuses the statistical service. During execution, receipts check source identity and integrity checking, ordered version identity, active dependencies and scope history. If $A_{s,j}$ is the current attachment candidate and $M_{s,j}$ the persistent history state,

$$M_{s,j+1} = M_{s,j} \wedge \mathrm{validAttempt}_{s,j}, \qquad A^{\mathrm{issued}}_{s,j} = A_{s,j} \wedge M_{s,j+1}. \tag{11}$$

Reconstructing a plan retains this source-scoped latch. Changed dependencies revoke affected cards; unrelated cards remain usable. The attachment observer modifies claim state while preserving canonical numerical decisions, installation ordering and modeled work. Its separately measured CPU and the noninterference proof are provided in the supplement.

4. Changing inspection requirements

4.1. Acquisition, association and report outputs

The image task connects a captured surface region to planned CAD addresses. Up to 15 face/barycentric targets are selected along the prescribed route before sensing. The camera model raycasts the airframe mesh, applies procedural surface texture, lighting, exposure blur and intensity noise, and rasterizes 25-mm-radius diagnostic fiducials. A nearest-palette decoder estimates their centers from the pixels. These correspondences provide an observable acquisition and CAD-association task with separate optical, geometric and timing requirements. Association uses the localization version installed at the sampled capture time $c \in \{0.1, 0.2, 0.3\}$ s.

At a planned target with range d, pixel focal length f_{px} and absolute incidence γ, geometric GSD is $g = d/(f_{\mathrm{px}}\gamma)$. We score the maximum GSD and blur over fixed planned addresses, acquired coverage from decoded landmarks, reprojection error in pixels, CAD-target displacement and positive-version readiness. At speed v and exposure τ, the target blur proxy is $\max_i v\tau/g_i$; the raster uses a Gaussian exposure surrogate

Table 3: Reference contract procedure. Statistical commitment precedes sensing; ordered numerical execution and evidence attachment retain distinct state.

Step	Operation
1	Compile the task into required events, engineering units and dependency projections.
2	Match the population, service and effective policy against the applicable card inventory.
3	Validate each card and its containment witness; retain candidates whose required bounds satisfy $U \le \delta$.
4	Minimize $\overline{L} + \lambda_{\mathrm{sel}}\overline{W}$ and commit the selected policy and evidence before opening the source. Return `REFUSED` for an empty catalogue.
5	Execute the retained solve under the stage reservation; publish eligible versions in source order.
6	Replace the active source contribution, replay acquired motion, and check source/version identity, dependencies and persistent history.
7	Expose the task output and its applicable evidence status; withdraw affected statements after a detected invalid history or dependency change.

characterized by its pixel-median GSD. For nominal CAD target q_i and its physical counterpart $q_i^\star$, the metre-valued association quantity is

$$a_{\mathrm{assoc}} = \max_{i \in \mathcal{I}_c} \left\| \widehat{m}_R q_i + \widehat{m}_t - (m_R^\star q_i^\star + m_t^\star) \right\|, \tag{12}$$

where $\mathcal{I}_c$ contains visible in-ROI targets with finite decoded centers and positive predicted depth. Reprojection separately compares pixels predicted from the active joint state with decoded centers. These scored quantities retain distinct units and dependencies. Full acquisition requires GSD at most 20 mm/pixel, blur at most 0.5 pixel and coverage at least 0.50. Full association requires coverage at least 0.50, reprojection at most 30 pixels, displacement at most 0.10 m and readiness at capture.

Full reporting adds delay at most 2 s and image age at most 1.8 s. With the fixed virtual raster/decoder work w_{camera} and load α,

$$t_{\mathrm{report}} = \max\{t_{\text{first positive}}, c + 0.05 + \alpha w_{\mathrm{camera}}\}, \qquad a_{\mathrm{image}} = t_{\mathrm{report}} - c. \tag{13}$$

The virtual producer clock makes processing load and delivery feasibility explicit. The thresholds define the acquisition, association and reporting services evaluated here. Their complete eight-variant specification and all unit checks are supplied in the supplement.

4.2. Requirement containment and producer dependencies

A requirement R compiles into admissible intervals and a failure event:

$$S_R = \{z : \ell_{R,j} \le z_j \le u_{R,j},\ j \in J_R\}, \qquad E_{p,R} = \{z_p(X) \notin S_R\}. \tag{14}$$

Nonfinite required outputs fail; the compiler rejects invalid specifications and incompatible units. An anchor A covers R when $J_R \subseteq J_A$ and $\ell_{A,j} \ge \ell_{R,j}$, $u_{A,j} \le u_{R,j}$ for each requested quantity. The resulting witness establishes

$$S_A \subseteq S_R \quad \Longrightarrow \quad E_{p,R} \subseteq E_{p,A} \quad \Longrightarrow \quad \Pr(E_{p,R}) \le U_{p,A}. \tag{15}$$

On the simultaneous anchor-coverage event, this implication holds for the entire containment closure. Calibration-dependent anchor selection within that closure adds no separate event family. Dropping a required quantity defines a partial service and is recorded explicitly.

Producer changes require an additional condition. Let D_j be the fixed implemented dependency set of quantity j, and $D_R = \bigcup_{j \in J_R} D_j$. A dependency signature identifies the registered component snapshot projected onto D_R. Transfer requires the requested functions to remain identical on supported complete inputs:

$$z_{p',j}(X) = z_{p,j}(X) \quad (j \in J_R) \quad \Longrightarrow \quad E_{p',R} = E_{p,R}. \tag{16}$$

Together with containment and an unchanged population, this equality preserves the old anchor bound. Producer compatibility is evaluated against the trusted implemented dependency specification. The three rules serve distinct implementation decisions: containment authorizes a changed requirement, functional invariance retains evidence after an unrelated provider change, and source-history validity permits attachment. The following theorem composes these rules into the correctness property of the operating mechanism.

Theorem 1 (Coverage of an authorized event)**.** *For one fixed calibration family, consider a request committed before sensing. Let a valid anchor A cover its requested failure event E_R, with $U_A \le \delta$. Assume the deployment population is matched and the requested producer functions remain invariant. On the simultaneous calibration-confidence event,* $\Pr(E_R) \le \delta$*. For an issuance indicator I generated by the source-history monitor,*

$$\Pr(E_R \cap \{I = 1\}) \le \delta. \tag{17}$$

A detected invalid attempt makes issuance false for subsequent attachments in that source scope while preserving valid numerical installation.

Proof. Requested-function invariance preserves the request event, and the containment witness gives $E_R \subseteq E_A$. Simultaneous anchor coverage yields $\Pr(E_R) \le \Pr(E_A) \le U_A \le \delta$. Intersecting with the issuance event preserves the upper bound. The conjunction in Eq. (11) remains false after a failed attempt, independently of later numerical installations. □

The simultaneous confidence event has calibration probability at least $1 - \beta$. Equation (17) bounds the joint event of task failure and evidence issuance. The theorem justifies the implemented reuse and issuance decisions through the existing finite-family bound; its state property ensures that plan reconstruction preserves a detected invalid history.

A wider ROI changes acquired coverage, association outputs and camera work. At the same planned addresses, fixed pixel focal length and target geometry preserve geometric GSD. Optical evidence can therefore remain applicable while a complete image or report event requires renewal. Capture-time readiness similarly uses the version available at capture; future pose updates do not retroactively alter it. Policy and anchor commitment before sensing avoids per-image switching among marginally calibrated events.

4.3. Reuse with direct-renewal fallback

For a committed numerical policy π, let $\mathcal{A}_\pi(R)$ contain compatible anchors that cover R and satisfy its allowance, and let $\mathcal{D}_\pi(R)$ contain compatible exact-event cards. The evidence route is

$$Q_\pi(R) = \begin{cases} \text{reuse}(A), & \mathcal{A}_\pi(R) \neq \varnothing, \\ \text{direct}(D), & \mathcal{A}_\pi(R) = \varnothing,\ \mathcal{D}_\pi(R) \neq \varnothing, \\ \text{refuse}, & \text{otherwise.} \end{cases} \tag{18}$$

Both branches use the same simultaneous calibration event and are chosen before sensing. Reuse therefore adds a proof-construction option while the exact-event fallback retains directly supportable requests. The routing analysis fixes the original direct-renewal policy, separates support from proof-preparation work, and deduplicates proof identities by event, producer, population and calibration source manifest.

4.4. Fair evidence-reuse alternatives

Contract coverage uses explicit containment witnesses and dependency signatures. Complete procedural coverage independently implements the same interval and dependency semantics. Cached direct renewal evaluates predeclared requested events using applicable calibration records, including invariant pools available to the other methods. Exact-card matching is the limited anchor-identity control. All receive the same quantities, cards, confidence allocation, records, objective and request cache. We evaluate cold preparation separately from first-query validation and warm lookup. Their archived method identifiers and complete specifications are supplied in the supplement.

Table 4: Shared processing services and distinct virtual camera work. Stage budgets concern numerical sender work; total work includes additional common components.

Service	Load α (factor)	Deadline D (s)	Stage budget B (ms)
Relaxed	10	0.5	20
Standard	20	0.3	12
Tight	40	0.2	6
Contended	60	0.3	12
Virtual raster + decoder: A 62.5 ms; B 125 ms			

5. Experimental design

5.1. Simulation models and calibration/test splits

The numerical study samples aircraft geometry, scan noise, motion and prior uncertainty, then executes the complete policy on each input. It contains 8,192 calibration episodes, 4,096 held-out test episodes and 1,536 stress episodes. The nominal mixture includes B767 and C172; the unfitted airframe is excluded from predictor training and has a separate calibration population. Each population has 4,096 calibration and 2,048 test episodes. Forty-eight pilot inputs establish simulation settings. Policies, events and processing services are fixed before test evaluation.

The image study uses two ROI settings: 192×144 pixels in A and 288×216 in B, with focal length 1,800 pixels and exposure 6 ms. Each study contains 4,096 calibration, 2,048 test and 256 stress inputs. Invariant optical quantities can use independently identified calibration blocks from both studies; other events use the matched study. The evidence inventory contains 3,072 cards under a reserved family of $K = 4,608$, with total confidence error $\beta = 0.05$. The 6,144 policy/evidence plans are selected from calibration information before test evaluation.

Nominal inputs vary prior deviations over 0.015/0.03/0.06 m, noise multipliers 1/2 and shape levels 0/1/2. Stress cases introduce occlusion, range degeneracy, larger prior error or map warp. Policy and request evaluations on one input are paired. Calibration counts refer to complete independently generated inputs, rather than the number of policy executions. The complete numerical specifications are supplied with the reproducibility package.

5.2. Comparisons and measurement boundaries

Matched module interventions separate predictor decisions, time/profile predicates and attachment checks. Full localization selection is compared with a task-aware calibrated procedure using identical source/version checks. The primary 16 paired intervals cover two profiles, four contrasts and two metrics at standard service, using 32,000 source-bootstrap resamples. Pointwise and Bonferroni-adjusted percentile intervals are approximate sampling intervals; exact binomial bounds govern admission. Sensitivity, subgroups, Pareto means and stress remain descriptive.

The continuous-route study contains 36 inputs: three geometries, three operating conditions and four realizations, with 12 methods and 299 scan opportunities per 600-s route. Its CPU timer covers the online processing loop. Preloaded-record preparation, numerical processing and camera components use separately defined timing boundaries. Primary Windows thread-CPU measurements have 15.625-ms clock quantization; processing budgets use the characterized operation clock. Registration probes and the camera-change campaign run on Linux/Python 3.12 with one numerical thread on an AMD EPYC 9V74 host. Their native timing and the primary Windows measurements describe separate executions. Point-count probes use caps 96, 192 and 384, recording actual counts and three timing repetitions per cap.

The camera-change campaign uses five CAD capture templates spanning both image studies, nominal and unfitted geometry. For each template, 64 random configurations perturb camera translation by a zero-mean 0.04-m standard deviation and rotation by 0.001 rad, with focal length uniformly sampled over 1,600–2,000 pixels, exposure over 4–8 ms and brightness over 0.70–1.15. The baseline ROI is 384×288, expanding to 512×384. Geometry, rasterization, decoding and association are recomputed for each before/after pair

Table 5: Primary-allowance service support. Configured labels retain the fixed request catalogue; unique contexts deduplicate compiled event identities within population and service. Full tasks retain all original requirements.

Study	Method	Labels /192	Unique /160	Full /24
A	Contract coverage	24	24	0
A	Complete procedure	24	24	0
A	Cached direct renewal	30	30	0
A	Exact-card control	18	18	0
B	Contract coverage	26	23	1
B	Complete procedure	26	23	1
B	Cached direct renewal	32	29	1
B	Exact-card control	20	17	1

using common input and noise. Four declared updates alter ROI, focal length, exposure or decoding. Four conditional fault fixtures introduce ROI-to-focal, decoder-to-readiness, exposure-to-focal or ROI-to-clock coupling. Output differences are computed before consulting the declaration and are compared with its affected quantities at tolerance 10^{-10}.

Archived-card routing evaluates all 320 unique request contexts with the direct-renewal numerical policy fixed. It separates service support from the selected evidence-preparation workload. Implementation consistency is evaluated through 15,360 declaration-model decisions, source-history cases and invalid-record checks. Together, these designs connect population calibration, numerical producer changes and execution-state correctness.

6. Results

6.1. Service support under reuse and direct renewal

At allowance 0.15, anchor reuse supports 24/192 configured labels in A and 26/192 in B; direct renewal supports 30/192 and 32/192. Several variants compile to the same event. Counting unique $(\text{population}, \text{service}, \text{event})$ contexts gives 24/160 and 23/160 for reuse, versus 30/160 and 29/160 for renewal (Table 5). The label and unique-event counts therefore describe complementary aspects of service support. Figure 4 retains every label and the additional direct-renewal services.

The full-association primary plan for B/unfitted/relaxed uses FlatCART. It has calibration count $229/2,048$, upper bound 0.143973 and independent test count $103/1,024$. The admission margin is 0.006027. Full acquisition and reporting remain unsupported; other accepted requests are explicitly partial or component services.

6.2. Reuse-first routing and evidence-preparation work

The reuse-first route retains all 30/160 and 29/160 directly supported contexts in the two studies. With the numerical policies fixed, 46 of these 59 contexts use compatible anchors and 13 use direct-event fallback. The difference from the 47 anchor-only contexts occurs in one Study B nominal/relaxed partial-association request: anchor-only selection uses FlatCART, whereas direct renewal selects Earlier CART. Holding the latter policy fixed routes this request to direct renewal, giving 22 reused and seven renewed contexts in B. Its selected proof catalogue contains 30 unique cards and requires 63,488 event–record evaluations, compared with 35 cards and 73,728 evaluations for exact-event preparation: a 13.89% reduction without changing supported contexts (Fig. 5a,b). The count measures event evaluation when constructing the selected proof catalogue from available calibration records.

The original anchor-only comparison retains a different trade-off: 47 supported contexts versus 59 for direct renewal. On the 47 matched contexts and their fixed policies, it requires 49,152 rather than 61,440 record evaluations, a 20% reduction. Across the complete catalogues, preparation evaluates 2,490,368 rather than 5,373,952 records, 53.66% fewer, with narrower support. The measured cold-time difference in Table 6

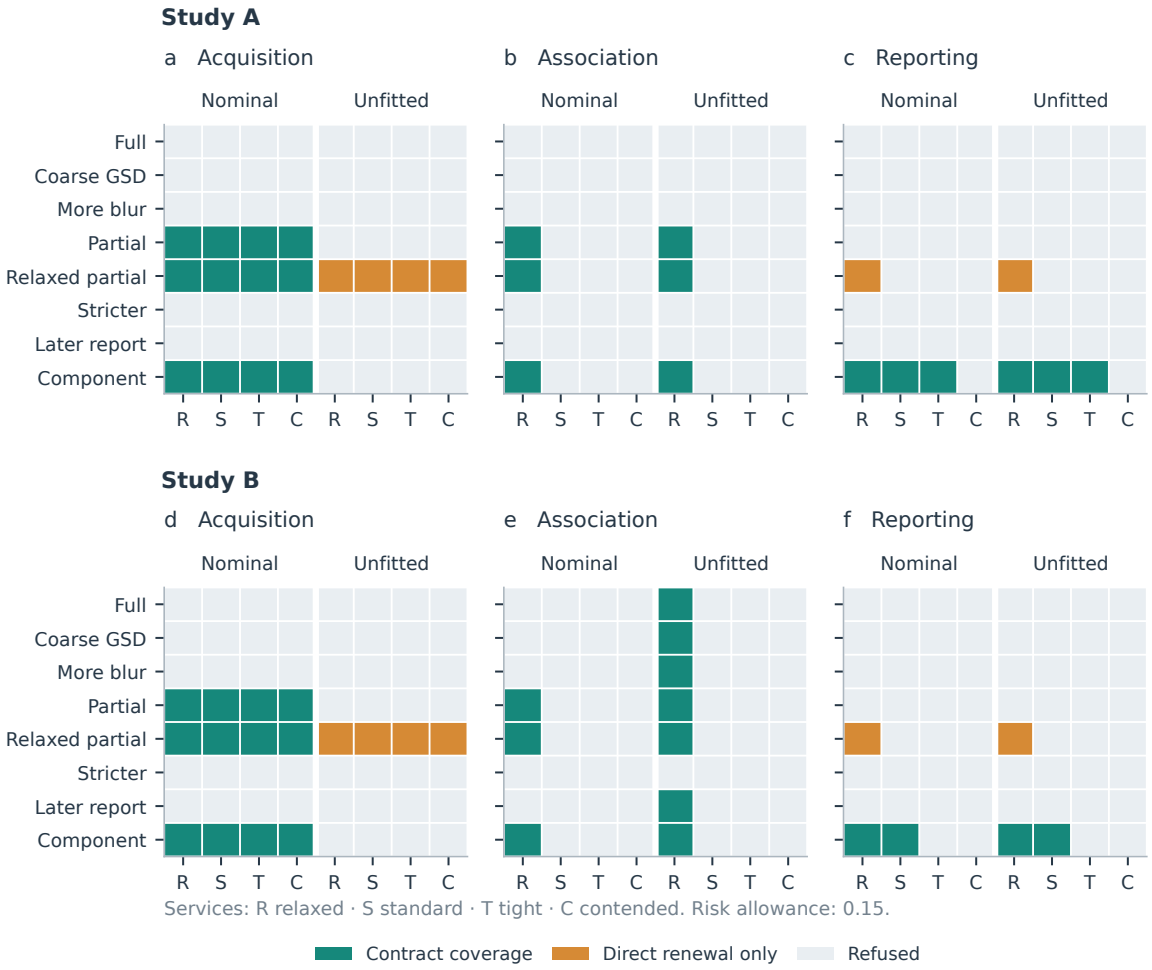


Figure 4: All 192 primary request labels per study. Teal cells show contract coverage; orange cells show additional cached direct-renewal support; grey cells are refused. Repeated event identities remain in this configured-label display and are separately deduplicated in Table 5.

Table 6: Preloaded-record timings from the original study. Cold time is the median of three trials; the first batch contains 384 queries. Warm lookup is reported in microseconds (μs), approximately 90–110 ns per cached dictionary lookup.

Method	Cards	Cold (s)	First batch (s)	Warm (μs)
Contract	1,152	0.179	6.408	0.091
Complete procedure	1,152	0.176	6.225	0.111
Cached direct renewal	2,560	0.365	9.991	0.098
Exact-card control	1,152	0.186	8.747	0.103

is 0.186 s. The absolute times describe lightweight event evaluation on preloaded records. Recurring warm lookups retrieve cached decisions, while direct fallback preserves the wider service coverage.

Multiplicity concerns calibrated risk functions; requirements in an anchor's containment closure introduce no additional tests. The primary comparison retains the originally reserved $K = 4,608$ for every method. An archived-count sensitivity groups tests using their event, producer, population and source-manifest signatures: 780 canonical anchor tests and 1,692 tests in the complete stored family. These are alternative

calibration-scope designs, each allocating its own total $\beta = 0.05$; a combined routed catalogue requires one allocation over its tested union. For the full-association example, the upper bound changes from 0.143973 to 0.140651 for the anchor design and 0.142133 for the complete design. At allowance 0.15, support remains 24/23 for anchors and 30/29 for exact events. Tighter multiplicity improves margins; it does not supply events outside the available closure. The sensitivity leaves all primary cards and plans unchanged.

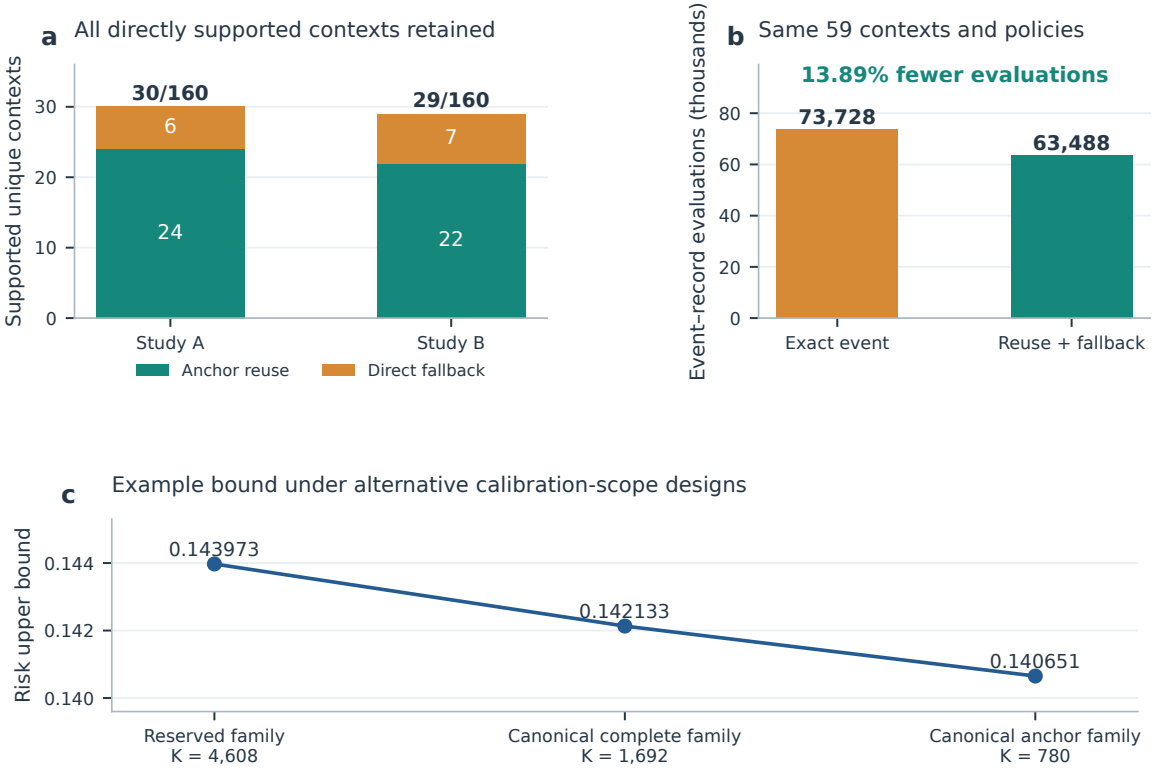


Figure 5: Evidence routing, proof preparation and calibration scope. (a) Reuse-first routing retains all 59 directly supported contexts: 46 reuse anchors and 13 use exact-event fallback. (b) With the same contexts and policies, selected proof preparation requires 63,488 versus 73,728 event–record evaluations, 13.89% fewer. (c) Alternative scope designs tighten the example risk bound while the allowance-0.15 support counts remain unchanged. Primary plans retain their reserved family.

6.3. Concrete source, dependency change and service recovery

An archived wide-ROI, unfitted-airframe source supplies a complete numerical and image case (Fig. 3). FlatCART installs versions one and two at 0.1 and 0.2 s; capture at 0.2 s consumes version two. All ten planned targets are acquired, reprojection is 6.04 pixels and CAD-target displacement is 19.4 mm. A prespecified consumer relaxation from 30 to 50 pixels remains covered by the original full-association anchor, reusing $U = 0.143973$ without recalibration.

Changing the decoder specification withdraws full association. A readiness-only request retains an applicable component bound $U = 0.056804$, whereas the optical component remains refused under its archived risk bound. Restoring the original decoder restores full association. This source individually passes the full-report thresholds, while the population-level full-report service remains uncertified: a favorable image does not authorize a missing population statement.

6.4. Producer changes across the camera operating range

The camera campaign recomputes eight task quantities for 2,560 paired updates across 320 configurations. ROI, focal-length, exposure and decoder updates retain the declared distinction between affected and unaffected quantities: all 1,280 declared-update cases pass the regression gate. Focal-length edits affect optical sampling, exposure affects blur, and decoder changes affect acquisition and association while preserving capture-time readiness.

Conditional fault fixtures create 747 activated updates among 1,280 fault cases. Every activated update exposes an undeclared output change and is vetoed; inactive branches retain the declared update behavior. ROI–focal coupling is activated in 163 cases, decoder–readiness in 160, exposure–focal in 104, and ROI–clock in 320. Figure 6 shows the changed quantities and the conditional activation counts.

These decisions use actual output differences, rather than a second implementation of the dependency table. The gate identifies the specific missing relation: optical sampling for a focal-length coupling, readiness for a decoder coupling, or delivery quantities for a clock coupling. Correcting the registered producer specification identifies which evidence requires renewal.

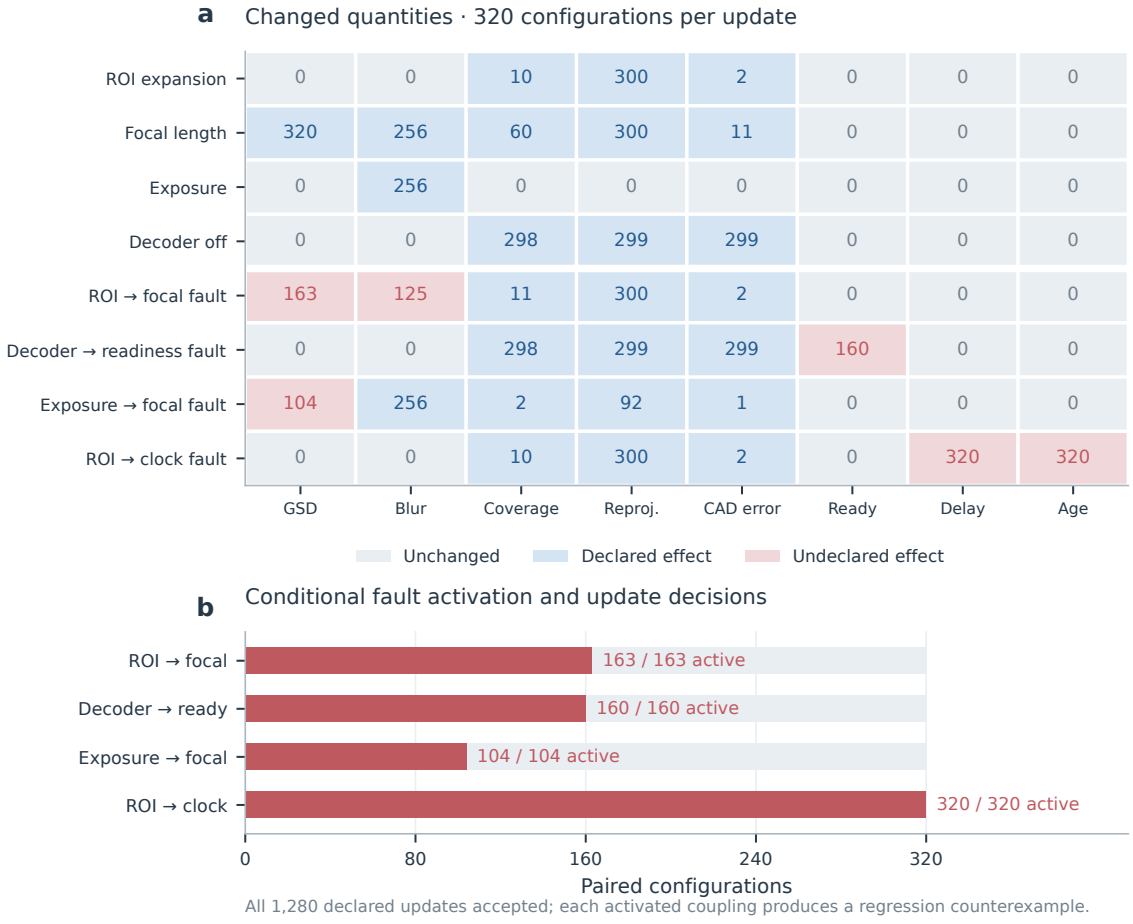


Figure 6: Executed camera-change campaign. (a) Quantities changed across 320 camera configurations per intervention; highlighted cells expose changes outside the declared provider projection. (b) Conditional fault activation and regression veto counts. Geometry, rasterization, decoding and task scoring are recomputed within each pair.

A separate declaration-model matrix evaluates 705 changes of initially admitted contexts: 489 withdraw

affected statements and 216 retain applicable statements. All restoration checks agree. The complete 15,360-decision matrix and invalid-record checks establish execution-state consistency alongside the numerical campaign.

6.5. ROI changes separate geometric quality from delivery feasibility

The excluded paired pilot in Fig. 7 uses the same source and landmarks for both ROIs. Across 48 paired pilots, fixed-target GSD is invariant: wider ROI changes acquired coverage and virtual production work while range, incidence and focal length remain fixed. The formal A/B test inputs are independent; their empirical differences are not a paired causal estimate. Optical projection invariance permits selective reuse, while affected coverage, association and report events require their own evidence.

The reporting clock also creates an analytical infeasible region. Equation (13) gives

$$a_{\text{image}} \geq 0.05 + \alpha w_{\text{camera}}. \tag{19}$$

This lower bound exceeds the full-task 1.8-s limit for A tight/contended and B standard/tight/contended: five of eight study/service combinations. Their reporting failure is built into the declared timing model. The other three combinations permit timing but can fail acquisition, association or actual delivery conditions. All full-report requests remain refused at the primary allowance.

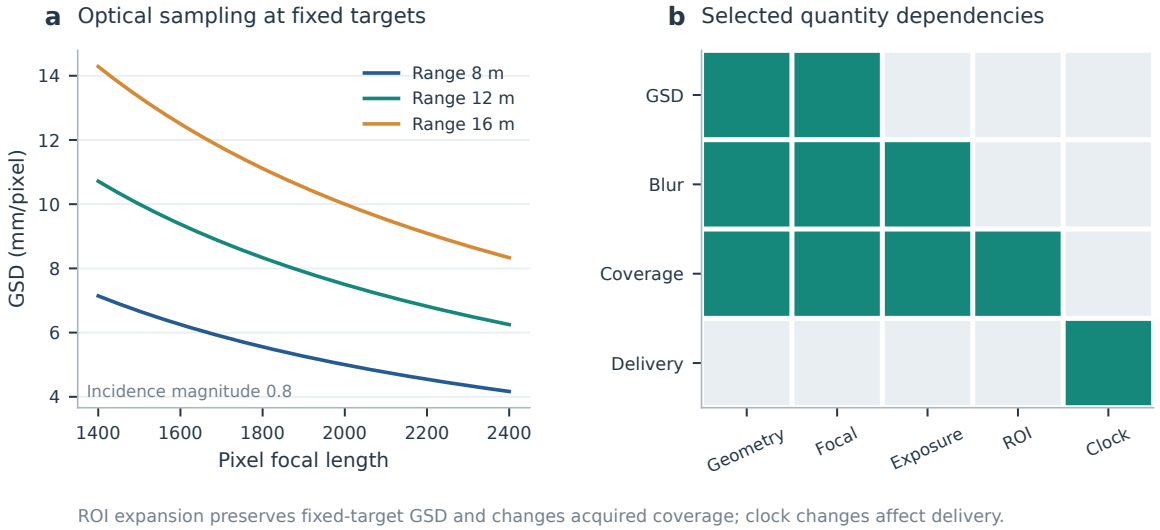


Figure 7: Optical sampling and request-specific dependencies. Geometric GSD follows range, focal length and incidence. Expanding ROI changes target coverage and image-processing work while preserving fixed-target GSD when those optical quantities remain fixed.

6.6. Localization accuracy, work and availability

At standard service, Contract-CART reduces integrated loss relative to Fixed2 by 15.68% on nominal geometry and 12.61% on unfitted geometry, with work increases of 16.09% and 16.07% (Table 7). Fixed4 and Final further reduce mean loss at greater work, while the RF policy remains close to Fixed2. Figure 8 displays all 22-policy means and the prespecified paired intervals, covering the four matched method contrasts.

At the primary allowance, Full admits 18 of 32 localization request contexts. Relaxed and standard services admit all four request types. Contention admits trajectory-only requests; tight service admits none. Refused requests remain in support denominators and retain the propagated prior during the corresponding execution.

Table 7: Standard-service test, $n = 2,048$ per population. L is integrated loss in 10^{-3} m^{2}s; W is total modeled work.

	Nominal			Unfitted		
Policy	L	W (ms)	Union fail (%)	L	W (ms)	Union fail (%)
Fixed2	13.833	9.221	8.64	19.938	9.220	10.06
Fixed4	11.295	12.109	7.52	17.350	11.736	9.28
Final	11.295	12.109	7.52	17.350	11.736	9.28
FlatCART	11.664	10.705	6.84	17.423	10.702	8.79
Contract-CART	11.664	10.705	6.84	17.423	10.702	8.79
FlatRF	13.844	9.973	8.64	19.936	9.953	10.06
ContractRF	13.844	9.973	8.64	19.936	9.953	10.06
Earlier CART	12.199	9.666	6.79	17.964	9.781	9.03
Full	12.301	9.721	7.13	17.964	9.781	9.03
Complete procedure	12.301	9.721	7.13	17.964	9.781	9.03

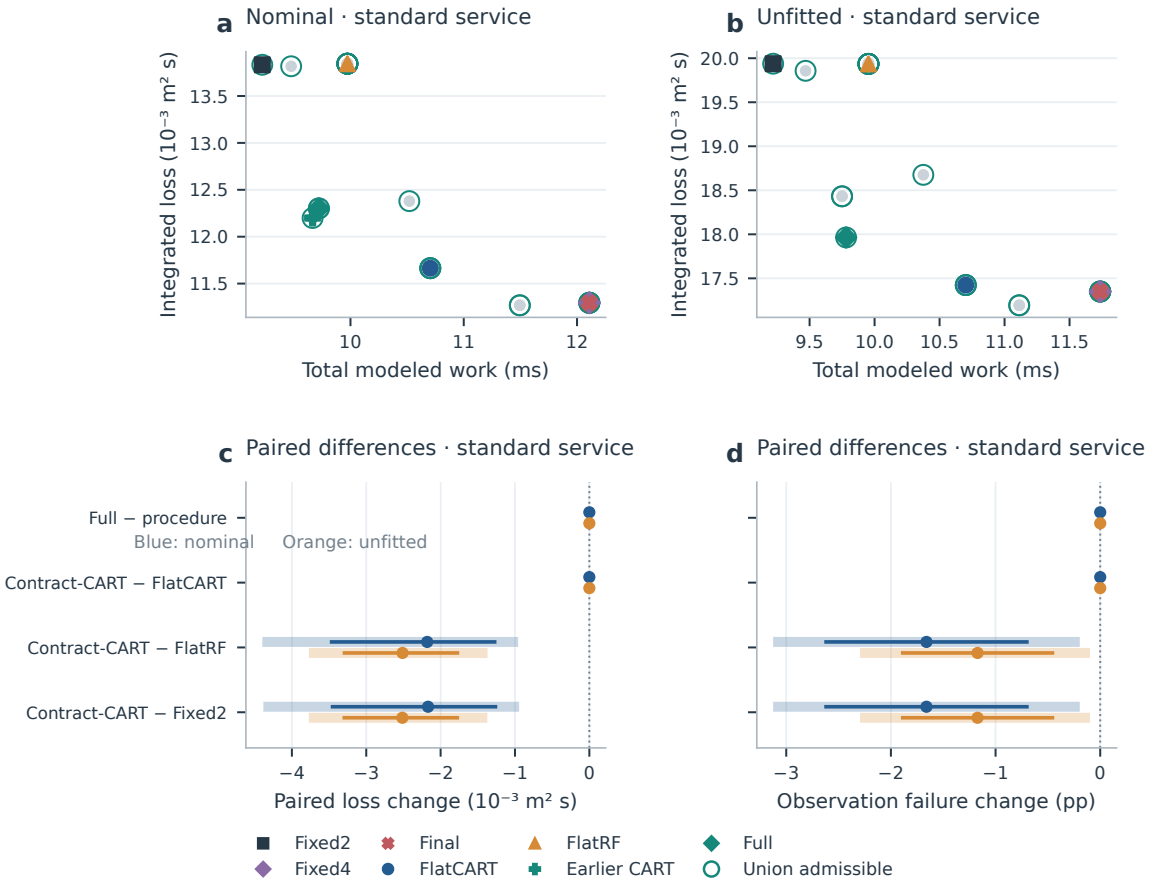


Figure 8: Standard-service localization. (a,b) All 22 policy means, highlighted controls and full selection; green rings mark direct-union eligibility at allowance 0.15. (c,d) All 16 paired intervals, colored by population: thin intervals are pointwise 95%, pale intervals are family-adjusted percentile intervals.

The gains also depend on episode difficulty. The point-count probe runs the same source solver at caps 96, 192 and 384, clipping to the available frontend cloud. Across the five inputs, actual counts range from 96

to 384 and native Linux CPU medians span approximately 6–18 ms. Source-pose errors are nonmonotonic in point count; the supplement reports each execution and acceptance decision. These probes characterize the reference solver and retain the primary 96-point policies.

The two largest nominal contributions account for 27.99% of the Contract-CART–Fixed2 net loss gain; the unfitted share is 10.02%. Diagnostic removal leaves positive mean gains of 0.001563 and 0.002265 m^2s, while the primary comparison retains every source. Complete tail and stress results are provided alongside these means.

6.7. Action predicates and evidence checks

Time screening changes every CART stopping endpoint under contention. On nominal test it reduces work by 3.600 ms and observation failure by 1.660 percentage points while increasing loss by 0.005094 m^2s. The unfitted changes are −3.583 ms, −1.904 percentage points and +0.004179 m^2s. The common next-stage reservation limits refinement under the tight budget. The supplement plots all-service time-predicate effects and the original attachment/dependency fixtures.

Evidence checks affect attachment correctness. Across 16 constructed invalid cases using actual fixed cards, stripped checks issue 16 incorrect attachments. Matching reduces this to four, current-version validation to two, and persistent source history to zero. The valid controls retain their attachments. Dependency edits selectively revoke cards: changing the RF forecast affects 168 of 528, whereas a shared clock change affects all 528. These fixed software fixtures measure compliance with the specified protocol.

6.8. Continuous-route accuracy and availability

Figure 9 shows all 12-method route summaries and highlights Fixed2, Contract-CART and Full. Ordinary routes often favor Fixed2 on CPU or accuracy; modest refinement opportunities provide limited return on forecasting and repeated publication. Under C172 contention, Contract-CART raises timely-correction success from 53.01% to 80.94%, with RMSE increasing from 5.237 to 5.341 cm. Full achieves 81.61% and 5.234 cm. Whole-online CPU is 2.121, 2.194 and 2.028 s, respectively, over each 600-s route.

Severe stress produces large route errors across the catalogue. The route selector refuses episode-to-route attachments and continues an explicit provisional recommendation. All 432 executions remain descriptive route evidence; their samples and policy switches are separate from episode calibration. Full tables and the predeclared continuous trace are supplied in the supplement.

7. Discussion

Evidence management connects the quality of an inspection output to the conditions under which its statistical statement was established. The connection becomes operational through three decisions. A changed requirement can reuse a stronger certified event. A changed component preserves evidence for quantities outside its implemented dependency projection. A changed numerical version preserves an output's provenance only when capture and execution ordering remain consistent. The typed representation makes these relationships available to the compiler and monitor as engineering operations.

The distinction between requirement change and producer change explains the role of reuse. Relaxing an association threshold changes the accepted event while retaining the measured quantity; containment supplies the existing bound. Changing focal length alters the quantity itself and therefore its applicable evidence. The camera campaign exercises these relationships through actual geometric and image computations, including conditionally activated dependencies. Offline regression identifies counterexamples to an update's declared effect, while registered signatures support lightweight runtime matching.

Direct renewal remains part of the same workflow. The fixed-policy route preserves all 59 directly supported contexts and uses anchors for 46, reducing selected proof-record evaluations by 13.89%. The additional request served through exact-event renewal illustrates why evidence selection must respect the numerical policy, rather than equating an event label with one interchangeable proof. Family-scope sensitivity tightens the bound but does not change support at allowance 0.15; the available event structure determines which requests can be served.

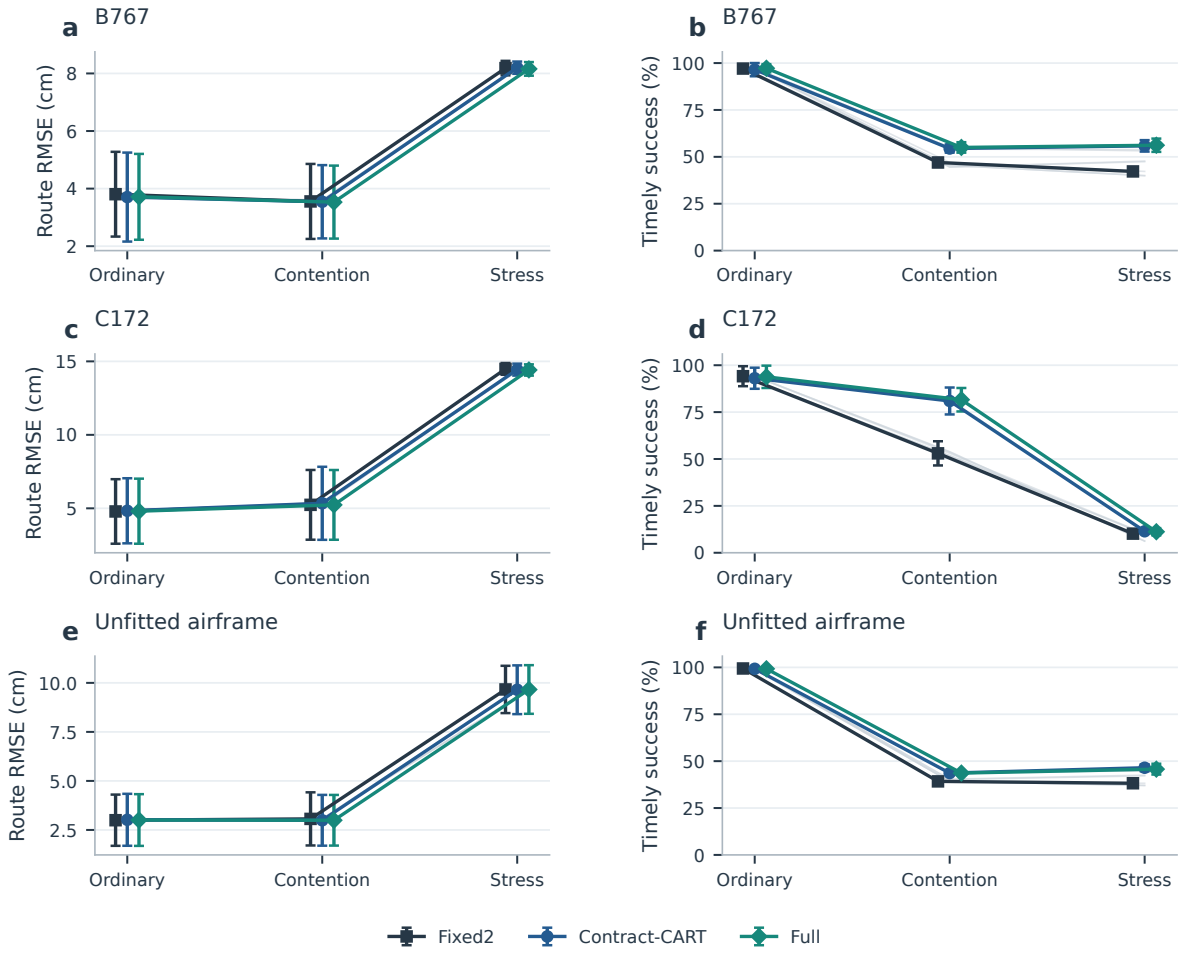


Figure 9: Complete-route results by geometry and condition. Grey traces retain all 12 methods; colored traces highlight Fixed2, Contract-CART and Full. Each point summarizes four matched 600-s inputs, with sample-standard-deviation bars. Route summaries carry no transferred episode bound.

The numerical and optical mechanisms identify actionable engineering choices. Processing reservations govern which estimate reaches capture. ROI governs acquired coverage; focal length and incidence govern fixed-target sampling. The reporting clock identifies a timing-infeasible request before a card is searched. These relationships allow an engineer to identify whether a request calls for a different processing service, a different producer or different calibration evidence. Numerical scheduling supplies the execution choices, and the evidence model governs their applicability through change. Appendix Appendix A records the complete implementation and evidence context.

8. Conclusion

Executable operating contracts manage statistical evidence across changes in CAD-guided inspection. Unit-typed events, producer dependencies and numerical versions govern retention, withdrawal and restoration. Reuse with exact-event renewal preserves all 59 directly supported contexts with 13.89% fewer selected proof-record evaluations. The 320-configuration camera campaign evaluates 2,560 paired updates, connecting the evidence rules to executed producer computations. Together, these operations provide a traceable workflow for localization, acquisition, association and reporting.

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Appendix A. Evidence scope and implementation context

Baseline equivalence and attribution. The standard-service Contract-CART numerical gains are shared by FlatCART; Fixed4/Final and FlatRF/ContractRF also have coincident reported means. FlatCART, time-only CART and Contract-CART coincide at relaxed/standard service. Profile predicates change none of the reported test outputs: the first thresholds lie outside ordinary noise/prior ranges, and the unfitted rule and common reservation limit refinement. Time-screen effects occur under contention. These observations do not establish global policy identity, so the 22-policy, $K = 528$ family remains fixed.

Full and the complete procedure reproduce numerical and execution-record outcomes, including 18/32 localization admissions. Contract and complete procedural coverage agree across 6,144 image-task plans, test decisions and declaration-model conformance cases; both share anchor accounting. These controls attribute numerical effects to scheduling and evidence effects to interval, dependency and history semantics.

Statistical statements and diagnostic verification. Population bounds assume iid complete calibration inputs, a fixed effective program, a matched deployment mixture and trusted dependencies. Theorem 1 bounds failure jointly with issuance; record-conditioned, subgroup and route risks require separate calibration. Producer transfer requires requested-function identity over the supported input domain.

The original five-source audit reproduces archived baseline pixels and quantities. The 320 camera configurations are random perturbations conditional on those five CAD/capture templates, not independently acquired aircraft scans. The four coupling classes are specified fault fixtures. Output differences falsify transfer on the tested path; regression agreement does not establish global functional identity or arbitrary-coupling completeness. The 705-edit matrix verifies declared-rule implementation. Trusted successor scopes recover using the same archived source identity. Family-scope sensitivity changes neither primary cards nor their allocation.

Physical implementation and measurement scope. The quantitative studies use synthetic localization, route and image inputs. Original photographs document project context. Physical implementation requires representative observations, platform timing and matched population calibration. The virtual operation clock, Windows component CPU, whole-route CPU and preloaded-record lookup times have distinct boundaries. Linux point-count medians use three repetitions per archived input and do not certify larger-point policies under the primary budget. Full reporting remains refused at the primary allowance; supporting it requires an appropriate producer service and calibration.

Data provenance and generative artwork. Main-text numerical results come from archived executed studies or the executed camera campaign. The campaign retains original population cards and policies; its configurations, seeds and all 20,480 quantity comparisons accompany the source package. Original diagnostic rasters and photographs are retained in the supplement. Optical curves evaluate the specified projection model.

Figure 1 is an OpenAI-generated conceptual illustration, providing context rather than a recorded deployment, experimental input or quantitative observation. OpenAI ChatGPT/Codex assisted code, analysis, writing and figure scripts. Quantitative plots use archived or recomputed values and deterministic graphics. This project reference manuscript identifies generated artwork separately from its experimental evidence.

Data and code availability. The repository `https://github.com/edrisskrah131-lang/A20-AEI-R8` provides the manuscript, supplement and editable package. It includes protocols, original solver/predictors, risk cards, version history, five complete replay inputs, campaign configurations, paired output tables and SHA-256 manifests. The reproduction script regenerates diagnostics, tables, figures and both PDFs. Full population source assets remain in the original versioned data release.